

with, informs Bhaskar. He says, "Some customers also prefer to have access to the source files of these graphical user interfaces (GUIs)/apps as reference for their own development."

For developing IoT applications

The rapid growth of Internet of Things (IoT) applications requires devices to be connected to the Web. Niraj Jha, CEO and MD, Fab.to.Lab, says, "Most people would like to have an Ethernet port or a Wi-Fi module built into the board." The biggest feature to consider for developing IoT-related products is inbuilt connectivity.

Mehta says, "It is important to consider the kind of connectivity options supported by the MCU like BLE4.0, 2.4GHz wireless technology and others."

Another important feature is the ease of programming while using the board. Programming becomes much easier if the board can run Linux or Android. "For smaller processors, I would choose a processor architecture that is well-supported and has tons of documentation available online," states Mehta.

She adds, "Companies like ST-Microelectronics and TI provide their own software development kits (SDKs), which wrap all internal hardware-level details for that particular MCU and make it easier to upgrade to a newer processor based on the same architecture."

Additional onboard features like flash-to-store program, lithium battery charger and SD card support are good to have.

"Last, power consumption and different sleep modes are essential. It is important to consider environmental factors such as adaptability of the board in remote locations running on solar power, or if it can survive on a small battery during unavailability of power for long durations."

Since the IoT is the talk of the town, Jha feels, some users might also want specific features like a display, motor/servo control for Internet-connected robotics applications or

Pricing trends

"Entry-level boards with prices starting from ₹ 800 are usually packaged with a limited set of onboard peripherals featuring 8-bit MCUs. Intermediate development boards range from ₹ 1500 to ₹ 5000 comprising 8-bit, 16-bit and 32-bit MCUs. There are full-featured development kits available for ₹ 5000 onwards featuring popular MCUs with a plethora of onboard peripherals and the ability to expand to new technologies using add-on modules."

—**Kanishka Shah, co-founder, Entesla**

"As pricing can range from a few hundreds to over ₹ 20,000, it is always good to do a thorough research before investing money in any of these development boards as you could stumble upon boards with similar specifications with a much lower price tag. However, some of these boards might be Chinese-made, which might compromise on quality and reliability. So, it is always a tough bargain between pricing, features and reliability."

—**Thushar V.P., technical director, Rhydo Technologies Pvt Ltd**

"Owing to the great demand for development boards and the increasing interest among the developer community, boards have seen a steep price decline over the past few years. Many sub-standard and cheap clones of these boards have also surfaced, which I would never recommend as these are usually unstable and fail over time."

—**Syed Anwaarullah, founder, PotentialLabs**

"Indian buyers are price-conscious and thus pricing seems to be at par for many boards available in the market. However, the trend is changing fast. Highly capable boards are being priced on value based outcomes rather than just bill of materials (BOM) costs and many buyers are appreciative of that."

—**T. Anand, managing director, Knewron**

"Pricing is a matter of concern for most developers today. Most mbed development boards cost ₹ 3000 or more and a majority of these need to be imported. Hopefully, prices will reduce in the future. Cheaper open source hardware is on the cards from many hardware communities. In fact, we at IoTBLR are exploring options on collaborative efforts towards designing some low-cost, open source MCU development kits for the Indian community. The variants will support mbed and Arduino integrated development environments (IDEs)."

—**Darshan Virupaksha, systems engineer, Altix Innovations, and co-organisr, IoTBLR**

"Pricing today starts from as low as the US\$ 9 full-featured, ultra-powerful, cloud-ready OAK boards to Java-ready, wirelessly-programmable US\$ 70 Tessel boards."

—**Niraj Jha, CEO and MD, Fab.to.Lab**

"TI development kits may be priced anywhere from as low as US\$ 9 to as high as US\$ 1000. Price point is something that is dictated solely by the kind of features the users are looking for in their development kits. In general, evaluation kits as launchpads tend to be of lower cost than full-blown development kits."

—**Nishant Bhaskar, embedded processing application engineer, Texas Instruments India**

sometimes a combination of a relay built into an IoT-ready board for home automation applications. He says, "Boards specifically made for home automation sometimes have other niche features like mesh networking built in. Other IoT boards may have features like secured cloud connectivity built-in into its software as well."

New technologies, prospects

ARM Cortex-M architecture based MCUs seem to be the future, feels Virupaksha. He says, "Getting used to ARM MCU architecture will be an added advantage for any career prospect." He adds, "Also, mbed today has revolutionised the development environment by making it an online

compiler. You just need a computer and an mbed board to get started."

As such no new revolutionary technology has emerged in the last few years, but according to Moore's Law, the speed is always increasing, notes Mehta. She says, "Intel has launched many new development boards like Galileo and Edison, which take the leverage of Intel's powerful and trusted platform."

Mehta adds, "Most technologies used are old like Wi-Fi but are packaged into newer processors and at highly reduced costs like the ESP8266 chip, which can act as a Wi-Fi solution with an integrated MCU. Older chips like AR9331 are packaged as easy-to-use modules like Onion